

Advanced SQL Questions

1. Which of the following SQL window functions computes the relative rank of a row within a window partition, leaving gaps in the ranking sequence when there are ties?

- A) ROW_NUMBER()
- B) DENSE_RANK()
- C) RANK()
- D) NTILE()

Answer: C

2. What is the primary advantage of using a Recursive Common Table Expression (CTE) in SQL?

- A) It improves the performance of simple aggregate queries.
- B) It allows querying hierarchical data like organizational charts.
- C) It prevents SQL injection attacks.
- D) It replaces the need for indexing tables.

Answer: B

3. In what order does the SQL engine logically evaluate the clauses of a standard SELECT statement?

- A) SELECT, FROM, WHERE, GROUP BY, HAVING, ORDER BY
- B) FROM, WHERE, GROUP BY, HAVING, SELECT, ORDER BY
- C) FROM, SELECT, WHERE, GROUP BY, HAVING, ORDER BY
- D) SELECT, WHERE, GROUP BY, HAVING, FROM, ORDER BY

Answer: B

4. When simulating a PIVOT operation using standard SQL (without a built-in PIVOT operator), which combination of functions is most commonly used?

- A) GROUP BY with CASE statements inside aggregate functions
- B) ORDER BY with string concatenation
- C) JOINS with window functions
- D) UNION ALL with subqueries

Answer: A

5. Which transaction isolation level prevents dirty reads, non-repeatable reads, and phantom reads?

- A) READ UNCOMMITTED
- B) READ COMMITTED
- C) REPEATABLE READ
- D) SERIALIZABLE

Answer: D

6. What happens if a subquery used with the IN operator returns NULL among its results?

- A) The query will throw a syntax error.
- B) The main query will ignore the NULL value and process the rest of the list normally.
- C) The main query will return no rows because NULL cannot be compared.
- D) It behaves identically to the EXISTS operator.

Answer: B

7. Which of the following is true about a Clustered Index?

- A) A table can have multiple clustered indexes.
- B) It dictates the physical order of data in the table.
- C) It only stores pointers to the data rows.
- D) It is typically slower for range queries compared to a non-clustered index.

Answer: B

8. In a CROSS JOIN between Table A (5 rows) and Table B (10 rows), how many rows will be present in the result set?

- A) 5
- B) 10
- C) 15
- D) 50

Answer: D

9. Which of the following commands is considered DML (Data Manipulation Language)?

- A) CREATE TABLE
- B) GRANT
- C) UPDATE
- D) DROP INDEX

Answer: C

10. What is the purpose of the COALESCE function in SQL?

- A) It concatenates multiple string columns into one.
- B) It returns the first non-null expression among its arguments.
- C) It merges two tables together similar to a FULL OUTER JOIN.
- D) It computes the statistical correlation between two numeric columns.

Answer: B